

Quiz 2 - Computational Physics I

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SCORE: 15/20

Date: Thursday 27 March 2025 Duration: 45 minutes

Credits: 20 points (5 questions) Type of evaluation: LAB

Provide short and concise answers to the following items:

1. (4 points) Systems of linear equations

- Explain how the Gauss elimination method for solving systems of linear equations works.
- List the main steps for solving such systems via symbolic algebra with SymPy in Python.

-1
a) Gauss elimination uses $\neq 0$ pivots to reduce the matrix into a upper triangular matrix, making the solution direct $\rightarrow$ To obtain a direct solution, the matrix should be a diagonal one.

b) Import library
↓
Define symbols $\rightarrow$ Ok, but how?
↓
Define system $\rightarrow$ How?
↓
Call the function? $\rightarrow$ Which one? $\rightarrow$ display

2. (4 points) Systems of nonlinear equations

- Indicate 2 methods that we can use to solve systems of linear equations in Python.

- 0.5
b) Briefly explain how each method works.

a) Graphical

We will get interceptions of the systems using ~~python~~ ^{python} indexing
`argwhere(-diff(-sign)).flatten()`

using the function for nonlin systems, and a guess on the neighborhoods of where the solutions are based on a plot, changing the neighborhoods gets us the different solutions

NumPy?

Which numpy method?

Which one? root?

3. (4 points) Scientific Data Formats

- List 2 types of scientific data formats used in physics.

- Briefly describe the features of each data format.

VTK

Can store large data sets;
uses groups, can be defined
in different structures, kind
of a tensor, can store different
kinds of data

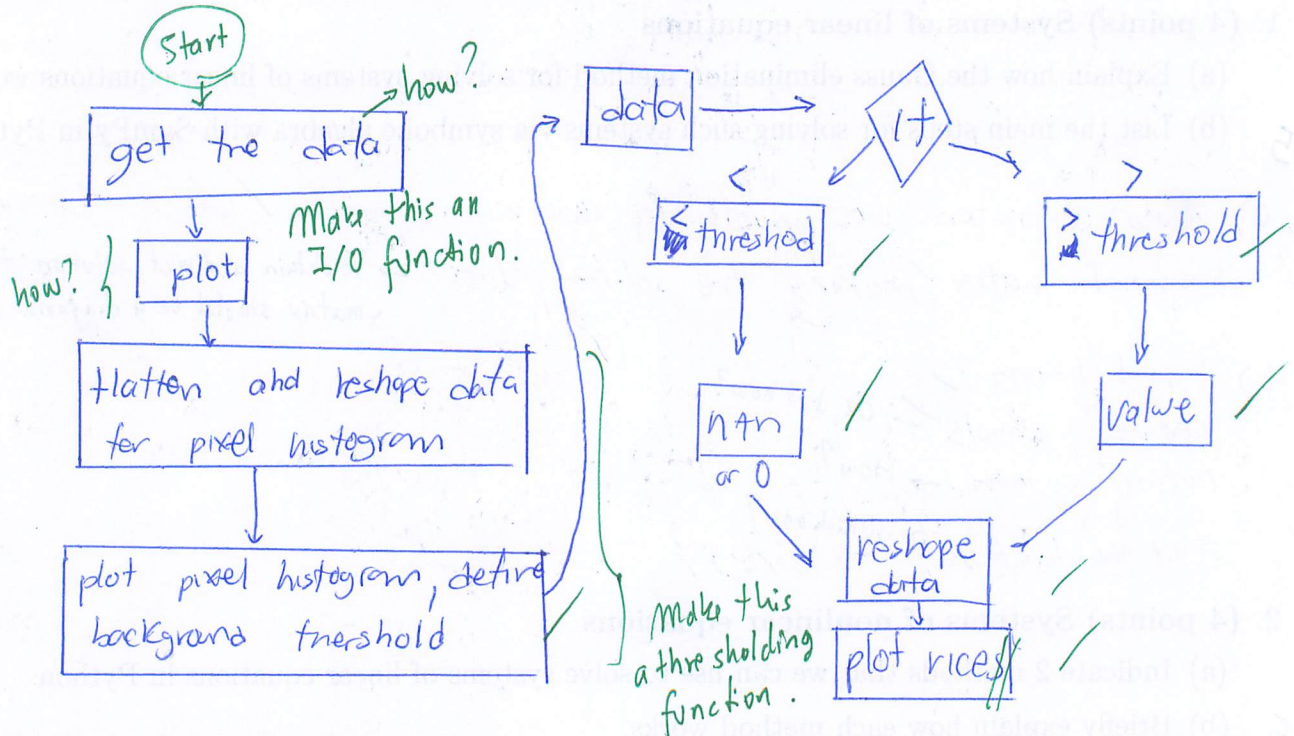
~~FFS~~ Binary

Refers to raw data, we have to know the data structure because this type of data has no header or metadata. Mostly needs reshaping

4. (4 points) Image processing



Imagine you obtain the photograph (shown on the left-hand side) of rice, and you are asked to isolate the rice grains from the background. Design and sketch a suitable algorithm workflow to achieve this goal in Python.



5. (4 points) Data processing

You have a dataset consisting of 50 HDF5 files, each containing time, gas density (ρ), and velocity components (v_x, v_y, v_z). Each file corresponds to a different time in the simulation.

- 2.5 (a) Write a Python function prototype for reading a single HDF5 file, computing the mean kinetic energy, and returning the time and the mean kinetic energy as NumPy arrays.
- (b) Sketch the loop needed to process all 50 files using the above function, so that you obtain and plot the mean kinetic energy versus time.

a) pseudo code should be commented.

```

def data(filename):
    """ Docstring? Use.
    Input: HDF5 lib h5py
    # Comment?
    open data = h5py.open(filename)
    time = data[0]
    rho = data[1]
    vel = data[2]  # vx, vy, vz.
    v2 = np.sum(vel, 2) ** 2
    K = 1/2 * rho * v2
    return np.array([t, K])
  
```

```


file_names = [...]
for n in range(50):
    t, K = data(file_names[n])
    data_t = [t]
    data_K = [K]
  
  
```

(b) is missing.